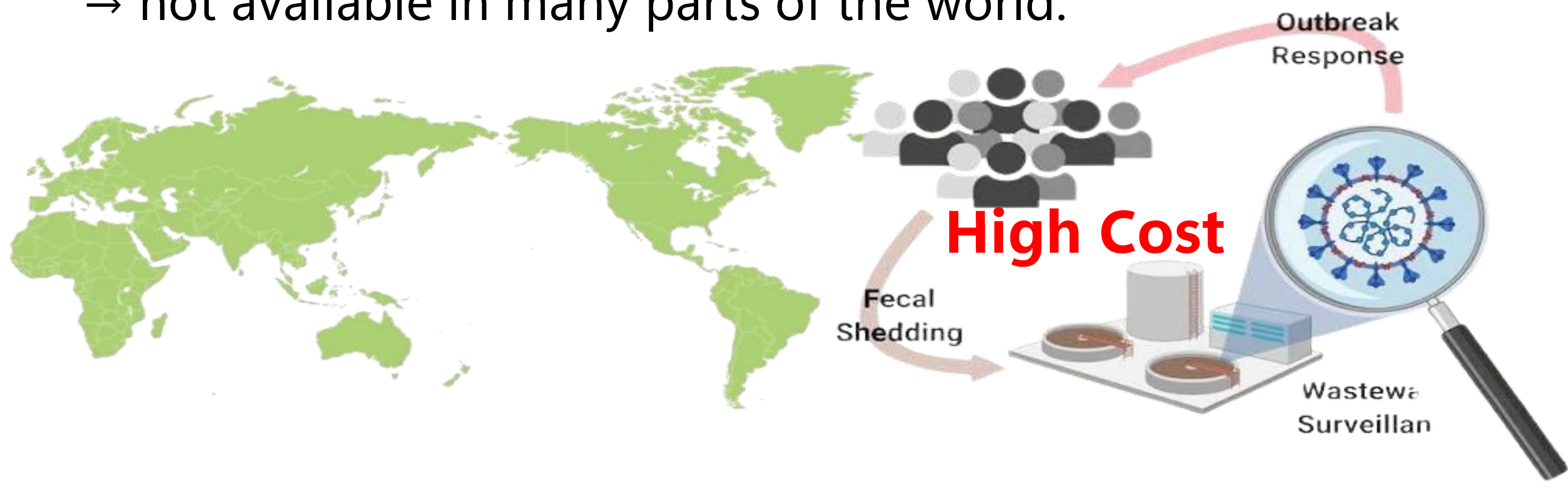


EpiSearch: Identifying Disease-Specific Leading Search Indicators Using Wastewater Surveillance Data in Korea

Motivation

- The risk of viral infection is strongly associated with the proportion of infected individuals in the community.
- Despite robust clinical testing infrastructure, **clinical data remain limited**.
- Wastewater-Based Epidemiology, **the high cost associated with advanced technology**.
→ not available in many parts of the world.



Internet Search Behavior

- When people are exposed to an infectious disease, they tend to search for information about the disease and its symptoms.

Keywords by Target Infectious Diseases

- SARS-CoV-2: 코로나, 기침, 발열, 호흡곤란
- Influenza: 인플루엔자, 독감, 플루, 고열, 근육통, 오한
- Norovirus: 노로바이러스, 장염, 식중독, 구토, 물설사, 복통

Search Term Combination Strategies

- Single Keywords
- Disease Name & Symptom Keyword Group
- Keyword Combination

Temporal Pearson Correlation

Temporal Correlation Analysis with WBE (Wastewater-Based Epidemiology) Pathogen Concentration Data.

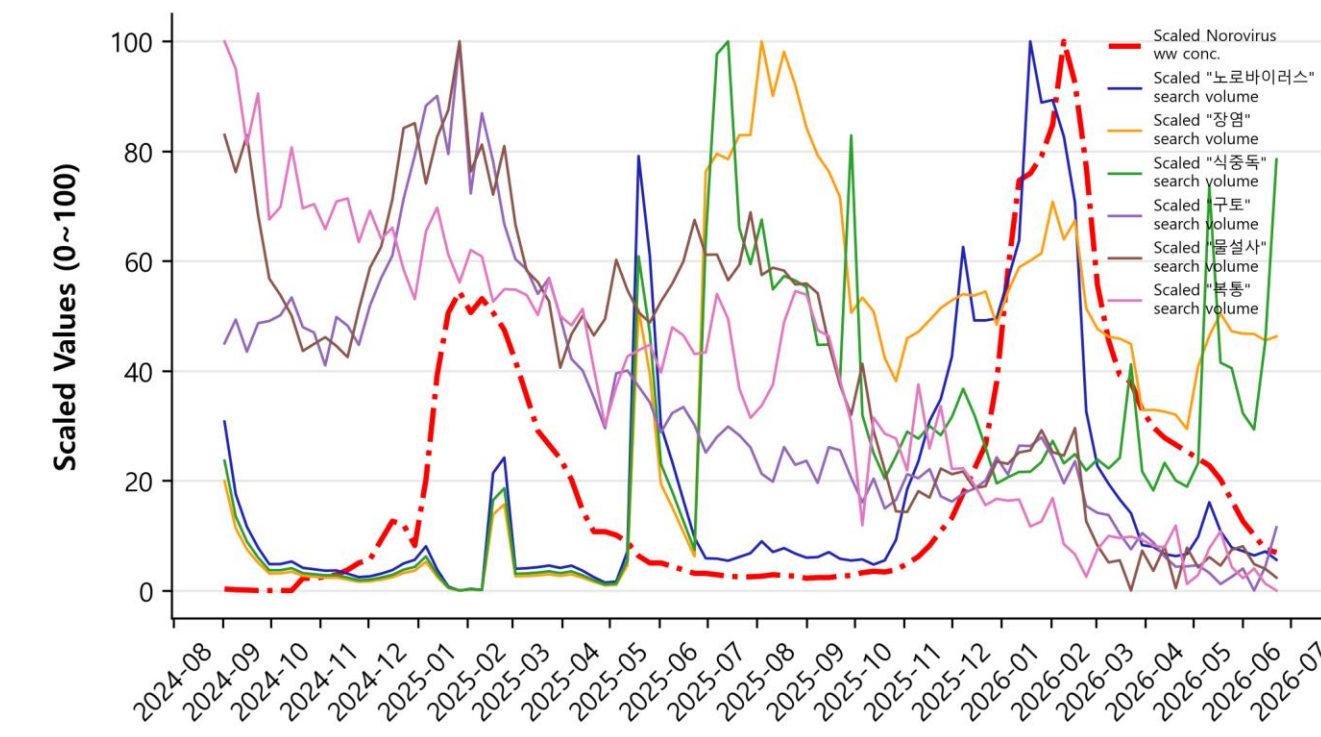
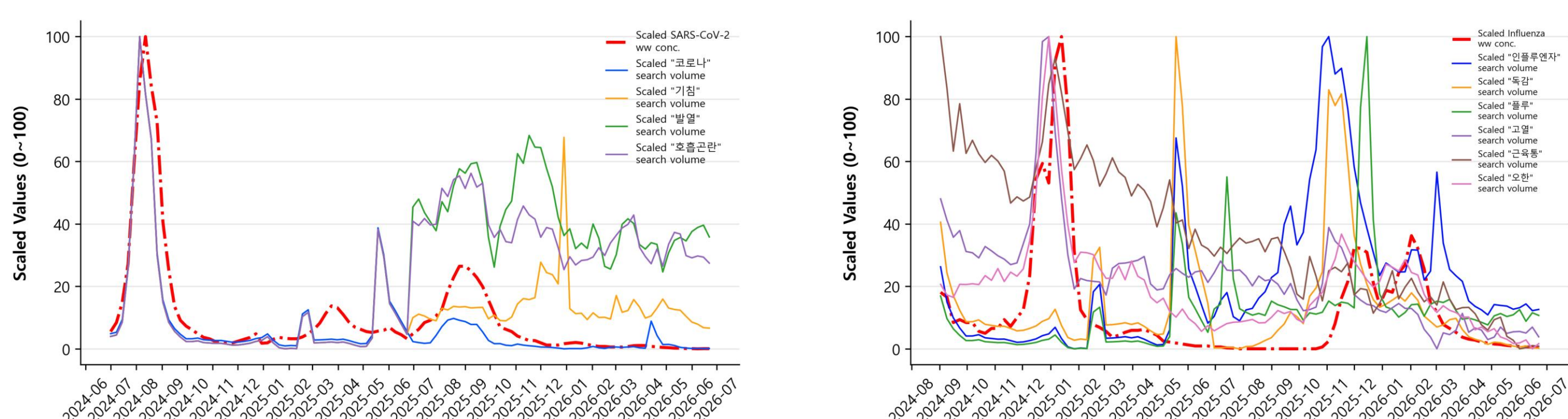
$$r_{xy} = \frac{\sum_{t=1}^n (X_t - \bar{X})(Y_t - \bar{Y})}{\sqrt{\sum_{t=1}^n (X_t - \bar{X})^2} \sqrt{\sum_{t=1}^n (Y_t - \bar{Y})^2}} \quad (1)$$

Time-Lag Analysis (Up to 3 Weeks)

By shifting the web search data and wastewater concentration data, respectively, and applying a time-delay correlation analysis, we identify the causal relationship.

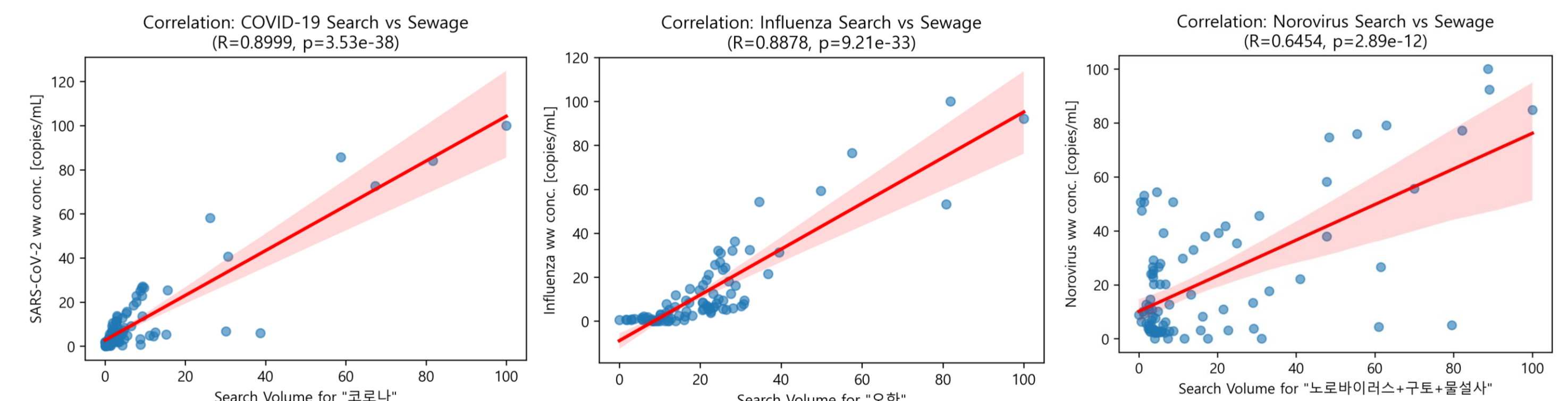
Data

- Concentrations of Infectious Diseases in Sewage** from Korea Disease Control and Prevention Agency (KDCA) Portal. 질병관리청
- Search Volumes for Keyword by Infectious Diseases** from NAVER Ads and NAVER Datalab. NAVER
- Min-max normalization from 0 to 100 is performed.
- Period: SARS-CoV-2 (Jul. 2024 – Jun. 2026) | Influenza & Norovirus (Sep. 2024 – Jun. 2026)

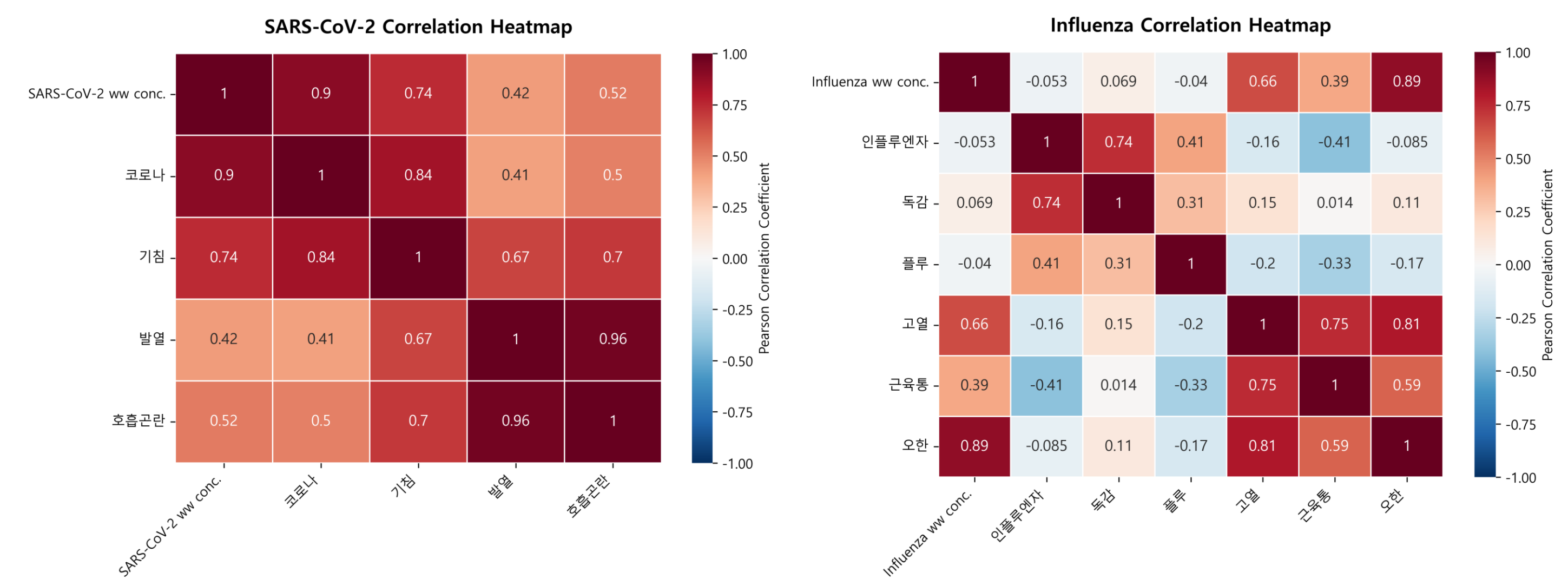


- Visualized the temporal trends of wastewater pathogen concentrations and keyword search volumes using graphs.

Results



- SARS-CoV-2 (r=0.9)** and **Influenza (r=0.89)** serve as leading indicators for **one week**, while **Norovirus (r=0.65)** serves as leading indicators for **two weeks**.
- Statistically significant correlations were confirmed for all of them.



- Identified the optimal keywords with the highest correlation coefficients through heatmap analysis.

Conclusion

The Need for Disease-Specific Strategies

It has been confirmed that web search term composition strategies and lead times vary depending on the type of infectious disease. **Proactive Decision Support via Early Warning Indicators** By overcoming the time-lag of WBE and clinical surveillance, validated web search combinations act as powerful early warning indicators. This approach is expected to serve as a crucial decision-support tool for the KDCA to optimize proactive responses and resource allocation.

Future Works

Expanded Variables & Scope Expansion

Enhancing the framework by incorporating diverse socio-environmental variables (e.g., meteorological data, population mobility) and expanding the scope to emerging infectious diseases with detailed city/provincial-level analyses.

Predictive Model & Early Warning System

Developing a real-time infectious disease monitoring Agent AI to support proactive decision-making by public health authorities.